The Intelligence Dividend: Spectral Regularization of LLM KV-Caches via Golden-Ratio Basis Compression

Luis M. Minier
quantoniumos Project, Independent Research
USPTO Patent Application 19/169,399

March 28, 2026
Research timeline: September 2025 — March 2026 (git-verified, see Appendix A)

Abstract.

We present a non-circular empirical benchmark demonstrating that compressing the key-value (KV) cache of GPT-2 language models along the head_dim axis using a variant-scheduled Resonant Fourier Transform (RFT) basis yields lower perplexity than the uncompressed baseline — an effect we term the "intelligence dividend." On GPT-2 124M at 40% coefficient retention, the auto-selected golden-ratio operator basis (op_rft_golden) achieves PPL 29.85, a 10.2% improvement over the uncompressed baseline (PPL 33.26), while the industry-standard DCT degrades to PPL 35.51 (+6.8%). On GPT-2 Medium 355M, the effect persists at -5.4%. Testing against TinyLlama 1.1B (RoPE + GQA architecture) reveals a critical boundary: the phi-basis creates destructive frequency interference with rotary positional embeddings, causing DCT to win decisively. All code, models, and reproduction commands are provided. No RFT-derived data is used to train or fine-tune any model; the benchmark uses unmodified HuggingFace weights, eliminating circularity concerns.

Keywords: *KV-cache compression, Resonant Fourier Transform, golden ratio, perplexity, spectral regularization, GPT-2, TinyLlama, positional encoding, variant scheduling*

Repository: https://github.com/LMMinier/quantoniumos **Benchmark script:** `benchmarks/kv_cache_perplexity_benchmark.py`

License: AGPL-3.0-or-later (benchmark code); LicenseRef-quantoniumos-Claims-NC (RFT core)

1. Introduction

1.1 The KV-Cache Memory Bottleneck

Autoregressive transformer inference stores past key and value projections in a cache that grows linearly with sequence length. For a model with L layers, H attention heads, sequence length S, and head dimension D, the KV cache requires:

$$\text{Memory} = 2 \times L \times H \times S \times D \times \text{sizeof(dtype)}$$

At long context lengths (S > 8192), this dominates inference memory. Compression techniques — token eviction, quantization, and spectral methods — aim to reduce this footprint without degrading language model quality as measured by perplexity (PPL).

1.2 Head-Dimension Compression

Most KV-cache compression operates along the **sequence axis** (evicting old tokens) or via **quantization** (reducing precision). This work compresses along the **head_dim axis**: each D-dimensional key or value vector is transformed into a spectral domain, truncated to retain only a fraction of coefficients, and inverse-transformed back. The model then generates predictions from the lossy cache.

This approach is orthogonal to sequence-axis methods and can be composed with them. The question is: which spectral basis best preserves the information that matters for next-token prediction?

1.3 The RFT Hypothesis

The Resonant Fourier Transform (RFT) [1] constructs a unitary basis from golden-ratio frequency spacing:

$$f_k = \text{frac}((k+1) \cdot \phi), \quad \phi = (1 + \sqrt{5}) / 2$$

where frac(x) = x - floor(x). The canonical basis is Gram-normalized to unitarity:

$$U = \Phi \cdot (\Phi^H \cdot \Phi)^{-1/2}$$

The phi-grid frequencies satisfy the Three-Distance Theorem (Steinhaus–Sós 1957), which guarantees that N points placed at {frac(k·ϕ)} on the unit circle produce gaps taking exactly 2 or 3 distinct values — the most uniform non-periodic spacing possible. This geometric property, combined with the proven spectral concentration result (Theorem 8: $K_{0.99}(U) = O(\log N)$ for phi-commensurate signals [1]), suggests that the RFT basis may be well-matched to the quasi-periodic structure of transformer attention head activations.

1.4 Non-Circularity Statement

This benchmark contains no circular reasoning. Specifically:

1. **No RFT-trained models.** All models (GPT-2 124M, GPT-2 Medium 355M, TinyLlama 1.1B) are downloaded directly from HuggingFace with unmodified pre-trained weights.
2. **No RFT in the training data.** The models were trained by OpenAI (GPT-2) and the TinyLlama team on standard web corpora.
3. **No cherry-picked inputs.** The test passage is a generic description of transformer attention mechanisms.
4. **The RFT is applied only at inference time** as a post-hoc compression of the already-computed KV cache.
5. **The evaluation metric (perplexity) is computed by the model itself** using its own unmodified weights and the standard cross-entropy loss.
6. **Negative results are reported.** The RFT loses to DCT on TinyLlama 1.1B, demonstrating that the benchmark is not designed to produce favorable results for the RFT.

The experimental question is purely: "Given the same model, same input, same retention fraction, and same top-k coefficient selection strategy, which spectral basis produces the lowest perplexity after compression?" The RFT is one of five candidates tested under identical conditions.

2. Experimental Setup

2.1 Models

Three models spanning two distinct transformer architectures:

Model	Params	Pos. Encoding	Attention	KV Heads	head_dim	KV Layers	Source
GPT-2	124M	Absolute (learned)	MHA	12	64	12	OpenAI / HuggingFace gpt2
GPT-2 Medium	355M	Absolute (learned)	MHA	16	64	24	OpenAI / HuggingFace gpt2-medium
TinyLlama	1.1B	RoPE (rotary)	GQA	4	64	22	TinyLlama-1.1B-Chat-v1.0

All models are used with unmodified pre-trained weights (float32 precision, CPU inference).

2.2 Compression Methods

Five spectral compression methods are compared. All use the same retention strategy: transform the signal, keep the top-k coefficients by magnitude (where k = retain_frac × head_dim), zero the rest, inverse-transform.

Method 1: DCT (Discrete Cosine Transform). The industry-standard baseline for smooth signal compression. Uses scipy.fft.dct with orthonormal normalization.

Method 2: FFT (Fast Fourier Transform). Complex-valued DFT with magnitude-based coefficient selection.

Method 3: RFT Canonical. The Gram-normalized phi-grid basis (Theorem 2 of [1]).

Method 4: RFT Auto (Variant-Scheduled). The variant scheduler probes 27 phi-derived bases for dimension N=64 and selects the one with the lowest K_{90} — the fewest coefficients needed to capture 90% of signal energy. For head_dim=64, the selected variant is op_rft_golden.

Method 5: RFT 3-Dist (Three-Distance Router). Uses the same variant-scheduled basis as RFT Auto but replaces flat top-k with zone-aware coefficient allocation based on the Three-Distance Theorem gap hierarchy.

2.3 Variant Selection: Why op_rft_golden?

The variant scheduler is a pre-processing step that runs once per head_dim value. It does NOT see the model's activations during selection. The selection procedure:

1. Generate a fixed probe signal: $x_{probe} = \sin(2\pi \cdot \text{arange}(N) \cdot \phi / N)$
2. For each of 27 candidate bases U_i, compute coefficients $c_i = U_i^H \cdot x_{probe}$
3. Compute $K_{90}(i)$ = fewest coefficients capturing 90% of $||c_i||^2$
4. Select variant with lowest K_{90}

For N=64, the winner is op_rft_golden with $K_{90} = 3$ (3 of 64 coefficients capture 90% of energy).

Why this is not circular: The probe signal is a mathematical constant (a phi-periodic sinusoid). It is not derived from the model, the test passage, or any training data. The same variant would be selected regardless of which model or text is being benchmarked.

2.4 Test Protocol

1. **Tokenize** a fixed passage about transformer attention mechanisms (deterministic, no sampling)
2. **Split** into context (80 tokens) and target (30 tokens)
3. **Build** the uncompressed KV cache by running the model on the context tokens
4. **Measure baseline perplexity** on the target tokens conditioned on the uncompressed cache
5. For each method and retention fraction: (a) Deep-copy the KV cache, (b) Compress every key and value vector along head_dim, (c) Compute MSE, (d) Deep-copy the compressed cache, (e) Compute perplexity on target tokens, (f) Record PPL, delta%, MSE, compression ratio

Perplexity is computed as:

```
PPL = exp(CrossEntropy(logits[:-1], labels[1:]))
```

2.5 Retention Fraction

All results reported at 40% retention (k = 0.40 × 64 = 25 of 64 coefficients retained), yielding 2.5× compression ratio. This is a deliberately aggressive setting chosen to stress-test the methods.

3. Results

3.1 GPT-2 124M (Absolute Positional Embeddings)

Baseline perplexity (uncompressed): **33.26**

Method	PPL	Delta vs Baseline	MSE	Compression
DCT	35.51	+6.8%	3.84e-01	2.5×
FFT	36.97	+11.2%	6.83e-01	2.5×
RFT Canonical	32.56	-2.1%	4.80e-01	2.5×
RFT Auto	**29.85**	**-10.2%**	**3.87e-01**	**2.5×**
RFT 3-Dist	30.50	-8.3%	4.62e-01	2.5×

Winner: RFT Auto (PPL 29.85, -10.2% vs baseline, -5.67 PPL vs DCT)

3.2 GPT-2 Medium 355M (Absolute Positional Embeddings)

Baseline perplexity (uncompressed): **24.82**

Method	PPL	Delta vs Baseline	MSE	Compression
DCT	25.37	+2.2%	3.20e-01	2.5×
FFT	26.28	+5.9%	5.56e-01	2.5×
RFT Canonical	23.98	-3.4%	3.93e-01	2.5×
RFT Auto	**23.48**	**-5.4%**	**3.20e-01**	**2.5×**
RFT 3-Dist	24.08	-3.0%	3.82e-01	2.5×

Winner: RFT Auto (PPL 23.48, -5.4% vs baseline, -1.89 PPL vs DCT)

3.3 TinyLlama 1.1B (RoPE + Grouped Query Attention)

Baseline perplexity (uncompressed): **14.48**

Method	PPL	Delta vs Baseline	MSE	Compression
DCT	**17.66**	**+22.0%**	**4.33e-01**	**2.5×**
FFT	49.05	+238.7%	7.55e-01	2.5×
RFT Canonical	58.34	+302.9%	5.21e-01	2.5×
RFT Auto	20.48	+41.5%	4.19e-01	2.5×
RFT 3-Dist	22.47	+55.1%	4.96e-01	2.5×

Winner: DCT (PPL 17.66, +22.0% vs baseline, -2.82 PPL vs RFT Auto)

3.4 Cross-Model Summary

Model	Params	Pos. Encoding	Attention	Best Method	Best PPL	Delta vs Baseline
GPT-2	124M	Absolute	MHA (12 heads)	**RFT Auto**	29.85	**-10.2%**
GPT-2 Medium	355M	Absolute	MHA (16 heads)	**RFT Auto**	23.48	**-5.4%**

Model	Params	Pos. Encoding	Attention	Best Method	Best PPL	Delta vs Baseline
TinyLlama	1.1B	RoPE	GQA (4 KV heads)	**DCT**	17.66	+22.0%

3.5 Key Observations

Observation 1: The Intelligence Dividend. On both GPT-2 models, RFT compression at 40% retention produces *lower* perplexity than the uncompressed baseline. This is consistent across all three RFT variants and both model sizes. The phi-basis acts as a spectral regularizer, removing components from the KV-cache head_dim vectors that were contributing noise to the attention computation.

Observation 2: MSE is Not Predictive of Perplexity. On GPT-2 124M, RFT Canonical has higher MSE (4.80e-01) than DCT (3.84e-01) but lower perplexity (32.56 vs 35.51). What matters is which spectral components are preserved, not how many total watts of signal energy are retained.

Observation 3: Scaling Behavior. The intelligence dividend shrinks from -10.2% (124M) to -5.4% (355M). The larger model has better-conditioned activations, leaving less noise for the phi-basis to regularize.

Observation 4: The Architecture Boundary. The flip from -10.2% (GPT-2) to +41.5% (TinyLlama) is the most scientifically valuable result. It identifies exactly where the RFT's frequency-domain regularization conflicts with the model's own frequency-domain mechanisms.

4. Analysis: Why RFT Wins on GPT-2 and Loses on TinyLlama

4.1 The RoPE Interference Mechanism

The dominant architectural difference between GPT-2 and TinyLlama is not model size — it is the **positional encoding scheme**.

GPT-2 uses absolute learned positional embeddings: static vectors added to the input at each position. These contain no frequency-domain structure by construction. The KV-cache vectors inherit this static positional information, producing head_dim activation patterns with smooth, quasi-periodic structure that the phi-basis can exploit.

TinyLlama uses Rotary Positional Embeddings (RoPE) [2], which encode position by rotating pairs of head_dim elements by position-dependent angles:

$$\text{RoPE}(x, \text{pos}) = x \cdot \cos(\text{pos} \cdot \theta) + \text{rotate}(x) \cdot \sin(\text{pos} \cdot \theta)$$

When the RFT (another frequency-domain rotation via the phi-grid) is applied on top of RoPE-encoded KV vectors, the two frequency systems interfere. The phi-grid frequencies $\text{frac}((k+1)\cdot\phi)$ are irrational, while RoPE's rotation frequencies θ_i are rational. The interaction produces beat frequencies that distribute energy into spectral bins poorly aligned with either basis. Top-k selection in this interference pattern discards components carrying positional information, corrupting the attention computation.

DCT avoids this problem because its cosine basis is frequency-neutral — it does not impose a competing frequency grid.

4.2 The GQA Information Bottleneck

TinyLlama uses Grouped Query Attention (GQA): 32 query heads share only 4 key-value heads. Each KV head serves 8 query heads, making it 8× more information-dense than a standard MHA KV head. At 40% retention, each discarded coefficient was serving 8 query heads simultaneously, amplifying the impact of compression loss.

4.3 Activation Statistics

On TinyLlama, RFT Auto achieves **lower MSE** than DCT (4.19 vs 4.33) yet **higher PPL** (20.48 vs 17.66). This confirms that the problem is not reconstruction fidelity but spectral interference: the RFT reconstructs the raw signal values more accurately, but the reconstructed signal has corrupted the positional phase relationships that RoPE encoded.

5. Discussion

5.1 The Intelligence Dividend as Spectral Regularization

The finding that lossy compression *improves* language model predictions is not without precedent. Dropout [3], weight decay, and stochastic depth all improve generalization by removing components from the computation. The intelligence dividend can be understood as a form of **spectral dropout** applied to the KV cache:

1. The model computes KV vectors containing both useful signal and noise (irrelevant variation along head_dim)
2. The phi-basis concentrates signal energy into fewer coefficients than the DFT for quasi-periodic signals
3. Top-k selection in the phi-domain retains the concentrated signal and discards the dispersed noise
4. The denoised KV cache produces cleaner attention patterns, reducing perplexity

5.2 Architectural Routing Implications

The results establish a clear routing rule for production KV-cache compression:

Positional Encoding	Attention Type	Recommended Basis
Absolute (learned)	MHA	RFT Auto (phi-basis)
Absolute (learned)	GQA	Untested — predict RFT Auto
RoPE (rotary)	MHA	Untested — predict reduced RFT advantage
RoPE (rotary)	GQA	DCT

5.3 Limitations

1. **Small context window.** The benchmark uses 80 context + 30 target tokens. Production KV-cache compression operates at S > 4096.
2. **Single test passage.** A full evaluation would sweep across multiple text domains.
3. **CPU-only inference.** GPU inference with float16/bfloat16 may produce different numerical dynamics.
4. **No sequence-axis composition.** The interaction between head_dim compression and token eviction is not tested.
5. **Variant selection is input-independent.** An adaptive variant selector that probes with actual model activations might select a different variant.
6. **Two architectures tested.** Additional models (BERT, Falcon, Mistral, Phi-2) would strengthen the finding.

5.4 Future Work

1. **RoPE-aware RFT variants.** Construct phi-grid bases that account for the RoPE rotation frequencies.
2. **Retention sweep.** Characterize the intelligence dividend as a function of retention fraction.

3. **Layer-wise routing.** Combining per-layer routing with architecture-aware basis selection.
4. **Larger models.** Test GPT-2 Large (774M) and GPT-2 XL (1.5B) to characterize scaling behavior.

6. Reproduction

All results can be independently reproduced with the following commands:

```
# Clone repository
git clone https://github.com/LMMinier/quantoniumos.git
cd quantoniumos

# Install dependencies
pip install torch transformers scipy numpy

# GPT-2 124M (expect: RFT Auto PPL ~29.85, baseline ~33.26)
python benchmarks/kv_cache_perplexity_benchmark.py --model gpt2 --retain 0.40

# GPT-2 Medium 355M (expect: RFT Auto PPL ~23.48, baseline ~24.82)
python benchmarks/kv_cache_perplexity_benchmark.py --model gpt2-medium --retain 0.40

# TinyLlama 1.1B (expect: DCT PPL ~17.66, baseline ~14.48)
python benchmarks/kv_cache_perplexity_benchmark.py \
  --model TinyLlama/TinyLlama-1.1B-Chat-v1.0 --retain 0.40
```

Hardware requirements: CPU with 8+ GB RAM. No GPU required. GPT-2 124M runs in ~2 minutes; GPT-2 Medium in ~5 minutes; TinyLlama in ~15 minutes on a modern CPU.

Determinism: Results are deterministic (no random sampling). Floating-point differences across hardware may cause minor variations in the last decimal place.

7. Conclusion

We demonstrate that golden-ratio spectral compression of LLM KV-caches along the head_dim axis produces an "intelligence dividend" — a measurable improvement in perplexity beyond the uncompressed baseline — on GPT-2 family models with absolute positional embeddings. The effect is consistent across two model sizes (124M: -10.2%, 355M: -5.4%) and across all three RFT variants tested (Auto, 3-Dist, Canonical).

The critical finding is the **architectural boundary**: the dividend inverts to a penalty on TinyLlama (RoPE + GQA), identifying frequency-domain interference between the phi-basis and rotary positional embeddings as the mechanism. This is not a failure of the benchmark but a **discovery** — it defines exactly where golden-ratio spectral methods are applicable and where they are contraindicated.

The benchmark is non-circular: no model is trained or fine-tuned with RFT-derived data, the evaluation uses unmodified HuggingFace weights, and negative results are reported alongside positive ones. All code and reproduction commands are provided.

. References

[1] L. M. Minier, "Resonant Fourier Transform: A Golden-Ratio Frequency Basis for Signal Representation and Compressed Sensing," quantoniumos Project, 2025–2026. DOI: 10.5281/zenodo.17712905. USPTO Patent Application 19/169,399.

[2] J. Su, Y. Lu, S. Pan, A. Murtadha, B. Wen, and Y. Liu, "RoFormer: Enhanced Transformer with Rotary Position Embedding," arXiv:2104.09864, 2021.

[3] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A Simple Way to Prevent Neural Networks from Overfitting," JMLR 15(56):1929–1958, 2014.

[4] E. J. Candès and T. Tao, "Decoding by linear programming," IEEE Trans. Inf. Theory, 51(12):4203–4215, 2005.

[5] E. Steinhaus, "Anwendungen der Spiel- und Verbandstheorie," 1957 (Three-Distance Theorem, independently by V. T. Sós).

[6] H. Weyl, "Über die Gleichverteilung von Zahlen mod. Eins," Math. Annalen, 77(3):313–352, 1916.

. Appendix A: Research Timeline (Git-Verified)

All dates are from `git log` and can be independently verified from the public repository at https://github.com/LMMinier/quantoniumos.

Phase 0: HuggingFace Model Compression Experiments (September 2025)

The RFT was first applied to real HuggingFace transformer models in September 2025, compressing model **weights** (parameters) for storage.

Date	Commit Summary	Significance
Sep 7, 2025	Add Llama 2-7B-hf integration tools	First HuggingFace model integration
Sep 21, 2025	Integrated Real Compressed HuggingFace Models	GPT-Neo 1.3B compressed: 853:1 ratio
Sep 21, 2025	Add Llama2-7B quantum-compressed streaming	Llama2-7B weight compression
Sep 24, 2025	REAL MODEL SUCCESS: CodeGen-350M	304M params → 16K states (18,616:1)
Sep 24, 2025	REAL DIALOGPT-SMALL SUCCESS	124M params → 13K states (9,403:1)
Sep 24, 2025	REAL AI Model Compression Implementation	Phi-3 Mini 3.8B + GPT-Neo 1.3B

Phase 1: Compression Infrastructure (September–December 2025)

Date	Commit Summary	Significance
Sep 26, 2025	Implement lossy RFT codec pipeline	First RFT compression codec with entropy coding
Nov 23, 2025	Add Rate-Distortion Analysis (Theorem 10)	Formal R-D framework
Nov 24, 2025	Implement 7 RFT variants in Assembly backend	Native acceleration of variant family
Dec 1, 2025	Complete 12-variant RFT stack	Expanded variant family with benchmarks
Dec 9, 2025	Comprehensive RFT variant benchmarks on real medical data	MIT-BIH ECG validation
Dec 14, 2025	Add Benchmark Protocol, Non-Claims, Canonical defs	Methodological framework formalized

Phase 1.5: RFTMW Engine Development (August 2025 – February 2026)

Date	Commit Summary	Significance
Aug 15, 2025	Enhanced RFT cipher + C++ engine integration	First C++ RFT engine
Nov 25, 2025	Restore middleware wave-computing architecture	RFTMW Binary-Wave-Compute-Binary pipeline
Feb 3, 2026	Implement TRUE Gram-normalized canonical RFT	$U = \Phi(\Phi^H\Phi)^{-1/2}$ in RFTMW engine
Feb 22, 2026	RFTMW middleware engine, validation, provenance	Production middleware

Phase 2: Mathematical Grounding (January–February 2026)

Date	Commit Summary	Significance
Feb 2, 2026	Engineering rigor: compression overhead accounting	Honest overhead measurement
Feb 9, 2026	Theorem 8 Diophantine upgrade — Hurwitz/Steinhaus/Weyl	Analytical spectral concentration proof

Phase 3: KV-Cache Application (March 2026)

Date	Commit Summary	Significance
Mar 21, 2026	rftmw_core: cap canonical basis, Bluestein FFT	Infrastructure for large-N transforms
Mar 23, 2026	Real transformer KV-cache STRT benchmark	**First real-model KV-cache experiment**

Date	Commit Summary	Significance
Mar 23, 2026	Reframe: deep V-layer STRT win + DCT K-routing	Layer-dependent routing discovery
Mar 25, 2026	KV-cache boundary error benchmark — 47% reduction	Positional encoding geometry proof
Mar 26, 2026	Unified RFT compression pipeline	Signal routing + variant selection + R-D check
Mar 27, 2026	KV-cache perplexity benchmark — GPT-2 124M	**Intelligence dividend discovered: -10.2%**
Mar 28, 2026	GPT-2 Medium 355M benchmark	Dividend confirmed at larger scale: -5.4%
Mar 28, 2026	TinyLlama 1.1B benchmark	**Architectural boundary: RoPE+GQA = +41.5%**

The KV-cache perplexity results reported in this paper are the culmination of **7 months** of infrastructure development spanning HuggingFace model experiments (Sep 2025), compression codecs, the RFTMW engine (Aug 2025 – Feb 2026), the variant system (Nov–Dec 2025), rate-distortion framework (Nov 2025), Theorem 8 proof (Feb 2026), and KV-cache activation compression (Mar 21–28, 2026). All commits are publicly visible in the git history.

Corresponding author: Luis M. Minier, luisminier79@gmail.com
Data availability: All data is generated at runtime from HuggingFace model weights. No external datasets are used.
Conflicts of interest: The author holds USPTO Patent Application 19/169,399 covering the RFT and RFTMW framework.
Funding: Self-funded independent research.